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Single-Dose Psilocybin Therapy for Alcohol Use Disorder: Pharmacokinetics, Feasibility, Safety, and Efficacy in an Open-Label Study

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Single-Dose Psilocybin Therapy for Alcohol Use Disorder: Pharmacokinetics, Feasibility, Safety, and Efficacy in an Open-Label Study

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Abstract

Background Psilocybin, a serotonin 2A receptor agonist with psychedelic properties, shows promise as a novel treatment for alcohol use disorder (AUD). While current studies involve two dosing sessions, the effects a single dose have not been investigated. **Aims** To investigate the pharmacokinetics, feasibility, safety, and efficacy of single-dose psilocybin therapy in AUD. **Methods** This open-label, single-group study investigated single-dose psilocybin therapy in ten treatment-seeking adults (eight men and two women; median age 44 years) with severe AUD. The treatment involved two preparation sessions, a high-dose psilocybin session (25 mg), and two integration sessions. Pharmacokinetics were determined by noncompartmental analysis, and changes in alcohol consumption, craving and self-efficacy, were assessed with a linear mixed model. **Results** Notable between-participant pharmacokinetic variations were observed, with peak plasma psilocin concentrations ranging from 14-59 µg/L. Alcohol consumption significantly decreased over the 12 weeks following psilocybin administration. Heavy drinking days were reduced by 37.5 percentage points (95% CI, -61.1, -13.9, p = 0.005), and drinks per day decreased by 3.4 units (95% CI: -6.5, -0.3), p = 0.035). This was corroborated by reports of rapid and sustained reductions in craving and increases in self-efficacy. **Conclusions** Despite pharmacokinetic variations, a single 25 mg psilocybin dose was safe and effective in reducing alcohol consumption in AUD patients. Larger randomised, placebo-controlled, single-dose AUD trials are warranted. **Funding** This work was supported by The Novo Nordisk Foundation (NNF19OC0058412), The Lundbeck Foundation (R-355-2020-945), The Health Foundation(21-B-0358) and The Ivan Nielsen Foundation. **Clinical trial registration:** NCT05347849

Keywords

alcohol use disorder, alcohol consumption, craving, psilocybin, single-dose, psilocybin therapy, plasma psilocin, pharmacokinetics

Introduction

In recent years, there has been growing interest in exploring the therapeutic potential of classic psychedelic drugs, which act as serotonin 2A (5-HT2A) receptor agonists, for treating several mental health disorders (Nutt et al., 2020). Among these, psilocybin, a naturally occurring compound found in certain species of mushrooms (Nichols, 2016), has shown promising effects in patients with a diagnosis of alcohol use disorder (AUD) (Zafar et al., 2023). This burgeoning interest is underpinned by early investigations into the therapeutic effects of LSD, a structurally similar compound, which hinted at the potential of psychedelics in AUD treatment (Chwelos et al., 1959). Although most studies from this early era lack modern scientific rigour, a contemporary meta-analysis of six randomised controlled trials (n=536) from 1966 to 1970 found long-term benefits of a single LSD administration on alcohol misuse and abstinence (Krebs and Johansen, 2012). Recent research has provided further support for the efficacy of psilocybin in AUD treatment (Bogenschutz et al., 2015; Bogenschutz et al., 2022; Heinzerling et al., 2023). Notably, these studies have consistently demonstrated efficacy using a regimen involving two dosing sessions. However, several clinical studies on depression have reported positive outcomes with just one dosing session (Griffiths et al., 2016; Ross et al., 2016; Goodwin et al., 2022; Raison et al., 2023; von Rotz et al., 2023). This underscores the need for evaluation of the efficacy of a single dose of psilocybin in the treatment of AUD.

Psilocybin can induce profound alterations in conscious experience, often mediating shifts in narrative structures of the self and the world (Griffiths et al., 2006). While the underlying therapeutic mechanisms are not fully understood, there is a growing body of evidence suggesting that these drug-induced experiences play a pivotal role in psilocybin's long-term therapeutic effects (Ko et al., 2022). Therefore, understanding what shapes these experiences is crucial to help advance its therapeutic applications. To this end, exploring the relationship between plasma concentrations of the active metabolite psilocin and the subjective experiences, as well as how they relate to long-term outcomes, is warranted. Presently, pharmacokinetics and pharmacodynamics studies are few and have been limited to healthy subjects (Hasler et al., 1997; Brown et al., 2017; Madsen et al., 2019; Becker et al., 2022; Holze et al., 2022; Ley et al., 2023; Holze et al., 2023), leaving a critical knowledge gap of psilocybin in patients with AUD. Since chronic alcohol consumption is frequently accompanied by perturbed metabolic profiles, which may lead to atypical drug responses and subsequently poor treatment response, the characterisation of the pharmacokinetic and dynamic properties of psilocybin is particularly important in the context of AUD. Highlighting this, earlier reports on psychedelic effects had indicated a potential insensitivity to psychedelics among individuals with AUD (Chwelos et al., 1959; Bogenschutz et al., 2015).

This open-label study is the first to assess the feasibility, safety, and efficacy of a single dose of psilocybin in patients with AUD, and to characterise its pharmacokinetic and pharmacodynamic properties.

Methods

Study design and overview

The study is an investigator-initiated, open-label, single-group, 12-week study involving individuals with AUD. Eligible participants were assigned to undergo a course of therapy delivered by two trained therapists, including a single fixed oral dose of 25 mg of psilocybin. Outcomes included feasibility, safety, pharmacokinetics and pharmacodynamics, as well as changes in alcohol consumption and related psychological outcomes. See Figure 1 for the study overview and timeline.

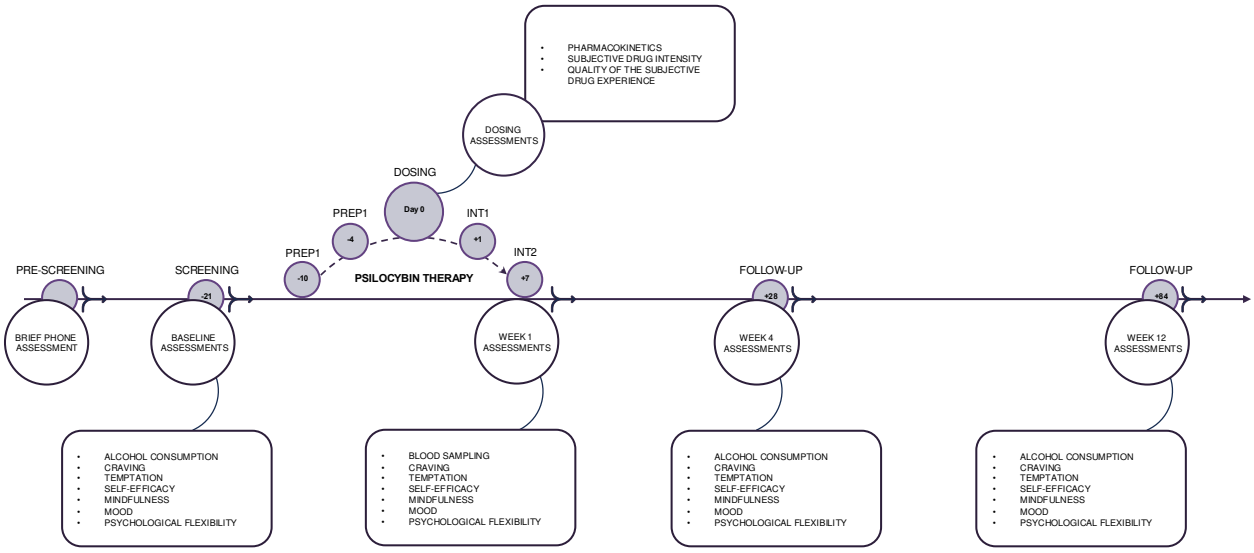


Figure 1 – Study overview. Days before and after Dosing Day (Day 0) are approximations. PREP1, the first pre-drug preparation session; PREP2, the second pre-drug preparation session; INT1, the first post-drug integration session; INT2, the second post-drug integration session.

Participants

Ten treatment-seeking adults (eight men and two women; median age 44 years (41 - 49); median body weight of 80.0 kg (74.2 - 84.2), were recruited from all regions of Denmark via our website www.alkoholforskning.dk/psilocybin. Enrollment began in March 2023 and ended in June 2023, with the last participant follow-up visit conducted in September 2023. A total of 37 individuals expressed interest in the study, of whom 15 underwent screening, and 10 were ultimately included. Participants were included if they met diagnostic criteria for AUD according to DSM-5 and alcohol dependence according to ICD-10, scored at least 15 on the alcohol use disorder identification test (AUDIT), reported at least five heavy drinking days,

defined as consuming 60/48 g (men/women) of alcohol or more on any day, in the past 28 days, and had a body weight of 60 - 95 kg. Exclusion criteria comprised a personal- or immediate family history of psychotic disorders, present suicidal ideation or a history of suicidal attempt, other drug use disorders (except nicotine dependence), ongoing treatment with medications for AUD (disulfiram, acamprosate, naltrexone, and nalmefene) or serotonergic medication, significant cardiovascular disease including prior myocardial infarction, angina pectoris, heart failure (NYHA class ≥ III), prolonged QTc interval (>450 ms/470 ms men/women) or uncontrolled hypertension (systolic blood pressure >165mm Hg, diastolic blood pressure >95 mm Hg). All participants received oral and written information and provided informed consent before being included in the study. Participants did not receive compensation, either monetary or otherwise.

Approvals and drug source

All procedures were approved by the Danish Ethics Committee of the Capital Region, Copenhagen, Denmark (H-20043832) and the Danish Medical Agency (2020-000829-55). The study was conducted according to the Declaration of Helsinki and was monitored by an independent study monitor (Good Clinical Practice unit, Copenhagen, Denmark). Trial registrations: ClinicalTrials.gov (NCT04718792), EudraCT (2020-000829-55). Filament Health Corp. provided the investigational medicinal product PEX010 (25 mg psilocybin capsules) for the study. PEX010 contains psilocybin, naturally extracted from Psilocybe cubensis mushroom fruiting bodies.

Psilocybin therapy

Psilocybin therapy, here defined as a therapy course of five consecutive sessions including two pre-drug preparation sessions, one psilocybin dosing session and two post-drug integration sessions, was delivered by a pair of trained study personnel (a psychotherapist and a medical doctor), consistent with previously suggested guidelines (Johnson et al., 2008). Overall, the course of therapy integrated elements from evidence-based psychotherapeutic interventions, including Motivational Interviewing (Miller and Rollnick, 2012), Acceptance and Commitment Therapy (Hayes et al., 2006), and Guided Imagery and Music (GIM) (Bonny and Pahnke, 1972), as these approaches were believed to foster motivation, openness, psychological flexibility, and skills for navigating altered states of consciousness. All sessions took place at the Copenhagen University Clinic for Psychedelic Research. The course of therapy is outlined below. See Jensen et al. for further details (Jensen et al., 2022).

Pre-drug preparation sessions

The primary goal of the preparation sessions was to establish a therapeutic alliance and prepare the participant for the psilocybin dosing session. These two sessions included a personal psychological review of important life themes related to their drinking behavior, information about practicalities and procedures for the dosing day, psychoeducation about the effects of psilocybin, and an experiential exercise. Further, it was important before the dosing day to establish agreements and demonstrate the use of therapeutic touch and physical support (hand-holding or hand on shoulder) for example, in case of distress. Ground rules also stipulated that participants could not leave the session except for escorted bathroom visits. To assist the participants in clarifying their life values and priorities, therapists utilised the "Bull's Eye" values-clarification exercise (Lundgren et al., 2012). The experiential exercise consisted of a standardised exercise in altered state of consciousness informed by GIM (Bonny and Pahnke, 1972), lasting 30 minutes and comprising three successive steps: 1) guided relaxation without music, 2) guided imagery to selected music, and 3) freely associated imagery in dialogue with the therapists while listening to the selected music. Through this exercise, participants were exposed to a simulated dosing scenario, where they lay with their eyes closed, listened to in-house curated music, and were guided into a lightly altered state of consciousness by the therapists.

Dosing session

Participants arrived at the clinic at 9:00 AM after consuming a light, low-fat meal and abstaining from caffeine and alcohol for the past 24 hours. A medical staff member performed tests to ensure the participant's suitability for the session, including a clean breath alcohol content (BrAC) test, a clean urine drug test, and an assessment for withdrawal symptoms using the CIWA-R scale. A peripheral venous catheter was inserted into the cubital vein, and a blood pressure cuff was placed on the opposite arm. The participant was then greeted by the therapists and chaperoned into the psilocybin dosing session room. Before administering the psilocybin, the leading therapist reviewed key elements from the previous preparation sessions, briefly revisiting the participant's treatment intentions and principles for navigating altered states of consciousness. When ready, the participant ingested a capsule containing 25 mg of psilocybin with water, and a curated, standardised 360-minute music program was played through loudspeakers. Typically, 10-15 minutes after ingestion, the participant reclined in bed, closed their eyes and focused inwardly with an open and curious attitude. Throughout the session, the therapists maintained an empathetic, non-directive presence, offering support and comfort as needed. Most of the time, there was minimal interaction between the therapists and the participants. Subjective drug intensity (SDI) assessments were conducted every 20 minutes. During the onset of the psychoactive effects, support and monitoring were intensified. This was done to alleviate any

psychological distress, such as anxiety or confusion, by providing reassurance and, if previously consented in the preparation phase, offering physical contact like a hand on the participant's shoulder for solace and grounding. As the peak drug effects began to wane, participants often felt inclined to converse. The therapists would listen and then gently remind them to return their attention inward. This descending phase of the drug effects was seen as an opportunity to focus more deliberately on personally important topics.

Post-drug integration sessions

The integration sessions, conducted on the day following the psilocybin dosing session and again one week later, were designed to promote the psychological well-being of the participants and facilitate the integration of their dosing experience into the treatment of AUD and daily life. Therapists employed a holistic approach to explore the time since leaving the research facility, which included sharing experiences with individuals outside the research group, reflecting on post-experience behaviors, thoughts, and feelings, as well as assessing factors such as sleep, dreams, appetite, and residual drug effects. Utilising deep listening skills, i.e., fully and empathetically engaging with the participants' verbal and non-verbal communication, therapists encouraged participants to provide a comprehensive narrative of their experience. They posed questions to promote presence and meaning-making, allowing participants to fully explore and articulate their experiences. Participants were also encouraged to reflect on the content of their experience, focusing on its relevance to their current life situation, motivations for change, and alcohol use. This reflective process aimed to integrate insights gained from the experience into the patient's ongoing therapeutic path.

Feasibility, cardiovascular effects, and adverse effects

The feasibility and safety were determined by the proportion of participants who completed all sessions of the psilocybin therapy and the rate and severity of adverse events, respectively. Blood pressure and heart rate were repeatedly measured in parallel with plasma psilocin concentrations.

Subjective effects of psilocybin

Subjective drug intensities were assessed every 20 minutes by the assisting therapist using a 0-10 Likert scale, with the question: 'How intense is your experience right now?' (0 = ‘Not at all’, 10 = ‘Very much’), throughout the dosing session. This approach is a meaningful summary measure of the psilocybin-induced experiential content (Stenbæk et al., 2021). Once the subjective effects subsided, as determined clinically and by two consecutive ratings of 0 intensity, participants were asked to complete questionnaires regarding the quality of the experience, including the revised Mystical Experience Questionnaire (MEQ30) (Hovmand et al., 2024a),

5-Dimensional Altered States of Consciousness Rating Scale (5-DASC) (Hovmand et al., 2024b), Ego-dissolution Inventory (EDI) (Nour et al., 2016), and Emotional Breakthrough Inventory (EBI) (Roseman et al., 2019).

Plasma psilocin concentrations

Blood samples were collected to assess plasma psilocin concentrations (PPC) before administration (control sample) and 40, 60, 80, 100, 120, 140, 180, 240, and 360 minutes after psilocybin administration. Samples of 6 ml were collected into lithium heparin tubes at room temperature and immediately centrifuged at 3500 rpm at 4°C. Plasma was briefly stored at -20°C during the dosing session and subsequently transferred to -70°C until analysis. Plasma concentrations of free, unconjugated psilocin were determined using ultra-high-performance liquid chromatography-tandem mass spectrometry, as described previously (Madsen et al., 2019). Analysis was conducted in units of µg/kg and converted to units of µg/L by multiplying by 1.024 kg/L plasma.

Clinical outcomes

Alcohol consumption, drinking patterns, and relevant psychological domains were assessed before and after psilocybin therapy to evaluate short- and long-term effects. Changes from baseline alcohol consumption were quantified at weeks 4 and 12 using the gold standard Timeline Followback Method (Sobell and Sobell, 1992), reviewing the past 28 days from the assessment day. Changes in drinking patterns were assessed at week 12 using the AUDIT test (Saunders et al., 1993). Changes in psychological outcomes were assessed at weeks 1, 4, and 12, including craving for alcohol measured by the Penn Alcohol Craving Scale (PACS) (Flannery et al., 1999) score; temptation and self-efficacy measured by the Alcohol Abstinence Self-efficacy scale (AASE) (DiClemente et al., 1994); mindfulness measured by the Mindful Attention Awareness Scale (MAAS) (Brown and Ryan, 2003); psychological flexibility measured by the Acceptance and Action Questionnaire (AAQ) (Bond et al., 2011); and depressive symptoms measured by the Major Depression Inventory (MDI) (Olsen et al., 2003).

Statistical analysis

We performed noncompartmental analysis calculations to determine the pharmacokinetic and pharmacodynamic parameters including peak PPC (C_{max}), time to reach Cmax (T_{max}), half-life ($T_{1/2}$), and the plasma psilocin concentration-time area under the curve from first to last sampling (AUCPPC) and the subjective drug intensity-time area under the curve (AUC_{SDI}). AUCs were determined using the trapezoidal-

up, log-trapezoidal-down approach. The relationship between PPC and SDI was assessed using a sigmoid Emax model: $\boldsymbol{SDI} = \frac{\mathbf{SDI_{max}} \times \mathbf{PPC}^{\boldsymbol{\gamma}}}{\mathbf{PPC_{50}} \times \mathbf{PPC}^{\boldsymbol{\gamma}}}$, where PPC_{50} is the PPC required to produce half of the maximum SDI, and γ is the Hill coefficient describing the steepness of the relationship. Lower and upper limits for the model were set to 0 and 10, respectively, to reflect the range of possible SDI ratings. Spearman's rank correlation coefficient was calculated to determine the strength and direction of the relationship between PPC and SDI. We used descriptive statistics (medians and interquartile ranges for continuous variables and percentages for categorical variables) for participant characteristics. The changes in continuous outcomes over time were evaluated using a linear mixed model. This model included follow-up times as fixed effects and assumed an unstructured covariance pattern to account for repeated measurements of the same participant. Effect sizes were estimated using Cohen's d, corrected for correlation between the repeated measurements. We calculated Spearman's rank correlation coefficient to explore whether certain baseline factors were correlated with plasma concentrations or subjective effects. In addition, we explored whether plasma psilocin concentrations and subjective drug effects were associated with changes in percentage heavy drinking days by including C_{max}, AUC_{PPC}, AUC_{SDI}, and MEQ30 total score as covariates in the linear mixed model. Further, we explored whether plasma psilocin concentrations and subjective drug effects differed between those who responded, defined as a reduction of > 50% in heavy drinking days from baseline to week 12, and those who did not, using a two-sample t-test. Missing data in the clinical outcomes were due to relapse and were consequently replaced with the participants' worst value observed. Missing PPC and SDIs were replaced with the last observation carried forward. Two-sided tests with a 5% level of statistical significance were applied. We used R software and the following packages: PKNCA, Drc, ggplot2, LMMStar.

Results

Participant demographics

Participants met 8 (7 - 10) of the AUD criteria, indicating severe AUD (defined as ≥ 6 symptoms out of 11), supported by an AUDIT score of 23.5 (20.2 - 27). Psychiatric comorbidities included self-reports of ADHD (one participant, unmedicated) and history of panic disorder (two participants). For seven of the ten participants, this was their first time undergoing any treatment for AUD. For detailed participant characteristics, please see Table S1 in the supplementary. For participant flow diagram, please see supplementary Figure S1.

Feasibility, cardiovascular effects, and adverse effects

All ten participants completed the psilocybin therapy course. A total of 46 adverse events were recorded, all categorised as mild, transient, and self-limiting, without requiring intervention (see Table S2). The most frequent adverse events were anxiety, headache, and fatigue and occurred within the first 24 hours of dosing. No participants required rescue medication during the study period. Within the first 100 minutes after administration, psilocybin moderately increased systolic blood pressure and only slightly the diastolic pressure and heart rate. See supplementary Figure S2.

Subjective drug effects

The SDI time courses of each participant are shown in Figure 2A. Corresponding pharmacodynamics are summarised in Table 1. All results from retrospective reports of the quality of the experience are summarised in Table S3. Following the oral administration of psilocybin, SDI peaked after 1.3 hours (1.30 - 1.93) and returned to baseline after 6.0 hours (4.40 - 6.30). The administered psilocybin dose of 25 mg induced intense and profoundly altered states of consciousness in most of the participants, as indicated by a median peak SDI of 10.0 (9.25 - 10.00) and by a median MEQ30 total score of 80% of maximum score (47 - 89).

Plasma psilocin concentrations

Plasma psilocin concentrations were determined from a total of 104 blood samples. Twice, we were unable to draw a blood sample at the pre-specified time due to catheter malfunction. Additionally, on six occasions, we took an extra blood sample outside the pre-specified times for comparison with the corresponding subjective drug response. The PPC time courses of each participant are shown in Figure 2B. Corresponding pharmacokinetics are summarized in Table 1.

The psilocybin was rapidly absorbed with a C_{max} of 24.0 µg/L (21.12 - 25.68) after 1.9 hours (1.40 - 2.23), and a half-life of 1.8 hours (1.60 - 2.07). The PPL-SDI relationship conformed well to the sigmoid E_{max} model, with a mean PPC_{50} of 10.6 (95% CI, 8.4, 12.9) µg/L and a Hill factor γ of 2.8. The goodness of fit was based on visual inspection (Figure 2C) and a lower residual standard error compared to a hyperbolic E_{max} model. Within participants, there was a strong relationship between PPC and SDI ($R_{Spearman}$ = 0.77, 95% CI, 0.68, 0.84, $p < .001$). Conversely, between participants, there was a considerable variability in the PPC and SDI time courses, clearly illustrated by the time profiles (Figure 2A-B) and indicated by a coefficient of variations for C_{max} of 47.6% and for AUC_{SDI} of 38.4%. Body weight, BMI, age, kidney and liver function did not correlate with plasma psilocin concentrations (C_{max}, AUC_{PPC}) and subjective drug effects (AUC_{SDI}) ($p > 0.05$) (Table S4).

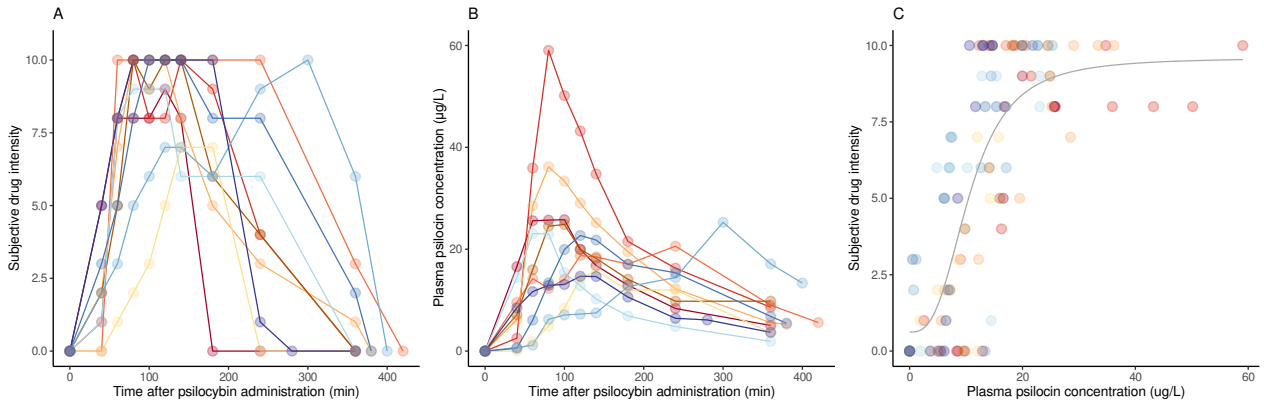


Figure 2 – Pharmacokinetic and pharmacodynamic profiles and their relationship. Concentration-time curves, effect-time curves, and the relationship between plasma psilocin concentration and subjective drug intensity are shown separately for each participant after the administration of 25 mg psilocybin. Of note, two participants stood out: one with a high C_{max} (red, participant no 2, table 1) and another with a delayed response (blue, participant no 8, table 1), but both showed no other notable differences from the rest. Data points are color-coded uniquely for each participant, and these colors are consistent across all Figures.

				Pharmacokinetics				Pharmacodynamics			
Participants (sex)	**Age**	**Body weight**	**mg/ body weight**	**C_{max} (µg/L)**	**T_{max} (h)**	**$T_{1/2}$ (h)**	**AUC_{PPC} (ug x h/L)**	**SDI_{max}**	**T_{SDImax} (h)**	**Return (h)**	**AUC_{SDI}**
1 (male)	45	82	0.3	25.8	1.7	2	76.8	9	2	3.7	14.8
2 (male)	46	78	0.32	59	1.3	2.3	131	10	1.3	6	26
3 (female)	35	78	0.32	20.6	4	NC	93.5	10	1	7	43.3
4 (male)	50	82	0.3	36.2	1.3	1.7	98.8	10	1.3	6.3	26.5
5 (male)	43	85	0.29	24.9	1.7	3.8	77.2	10	1.3	6	24.2
6 (male)	41	73	0.34	15.7	2.3	1.6	47.7	7	2.3	3.7	9.5
7 (male)	62	92	0.27	23.1	1	1.6	51.3	9	1.3	4.7	23.2
8 (female)	44	58	0.43	25.3	5	NC	80	10	5	6.7	38.3
9 (male)	37	92	0.27	22.7	2	1.6	76.8	10	1.7	6.3	36.8
10 (male)	68	73	0.34	14.7	2	1.9	51.5	10	1.3	4.3	27.3
median	44.5	80	0.3	24	1.9	1.8	77	10	1.3	6	26.3
(IQR)	(41-49)	(74.2-84.2)	(0.2-0.3)	(21.1-25.6)	(1.4-2.2)	(1.6-2.0)	(57.8-90.1)	(9.2-10.0)	(1.3-1.9)	(4.4 -6.3)	(23.4 -34.4)

Table 1 - Pharmacokinetics and pharmacodynamics of 25 mg psilocybin. C_{max}, peak plasma psilocin levels; T_{max}, time to reach peak plasma psilocin levels; $T_{1/2}$, half-life; AUC_{PPC}, area under the concentration-time curve from the first to the last sampling; SDI_{max}, maximum drug effect as measured by SDI ratings (0-10); T_{SDImax}, time to reach SDI_{max}; Return, time for drug effects to revert to baseline (SDI = 0); AUC_{SDI}, area under the effect-time curve. NC, not calculated (in two participants, $T_{1/2}$ could not be calculated due to a very delayed C_{max}).

Clinical outcomes

All clinical outcomes are summarised in Table 2. Individual and mean plots of participant changes over time are shown in Figure 3A-C. The percentage of heavy drinking days decreased from baseline to week 12 by -37.5% (95% CI, -61.1, -13.9, p = .005, Cohen's d = 1.02), and the drinks per day by -3.4 units (95% CI, -6.5, -0.3, p = .035, Cohen's d = 0.71). In line with this, participants reported a decrease from baseline to week 1 in

alcohol craving by -8.9 points (95 CI, -14.0 -3.7, p = .003, Cohen's d = 0.97), which was sustained at week 4 (-5.9, 95% CI, -10.56, -1.23, p = .018, Cohen's d = 0.77), and at week 12 (-5.1, 95% CI, -10.2, 0.06, p = .052, Cohen's d = 0.61). Further, participants reported an immediate change from baseline to week 1 in self-efficacy, i.e., confidence in the ability to abstain from alcohol by 17.6 points (95% CI, 4.3, 30.8, p = .015, Cohen's d = 2.39), which remained increased by week 4 (18.7, 95% CI, 9.44, 27.96, p = .001, Cohen's d = 2.72) and week 12 (18.5, 95% CI, 9.34, 27.65, p = .001, Cohen's d = 2.67). Trait mindfulness did not change after psilocybin therapy (baseline to week 12: -0.073 points, 95%, CI, -0.601, 0.455, p = 0.761, Cohen's d = 0.10). Neither did psychological flexibility (baseline to week 12: -4.1 points, 95% CI, -9.3, 1.1, p = 0.111, Cohen's d = 0.57) nor depressive symptoms (baseline to week 12: -0.8 points 95%, CI, -4.5, 2.9, p = 0.645, Cohen's d = 0.17). Individual analyses revealed that at week 4, nine of the 10 participants significantly reduced drinks per day, and at 12 weeks, seven of the 10 participants still reported significantly reduced drinks per day.

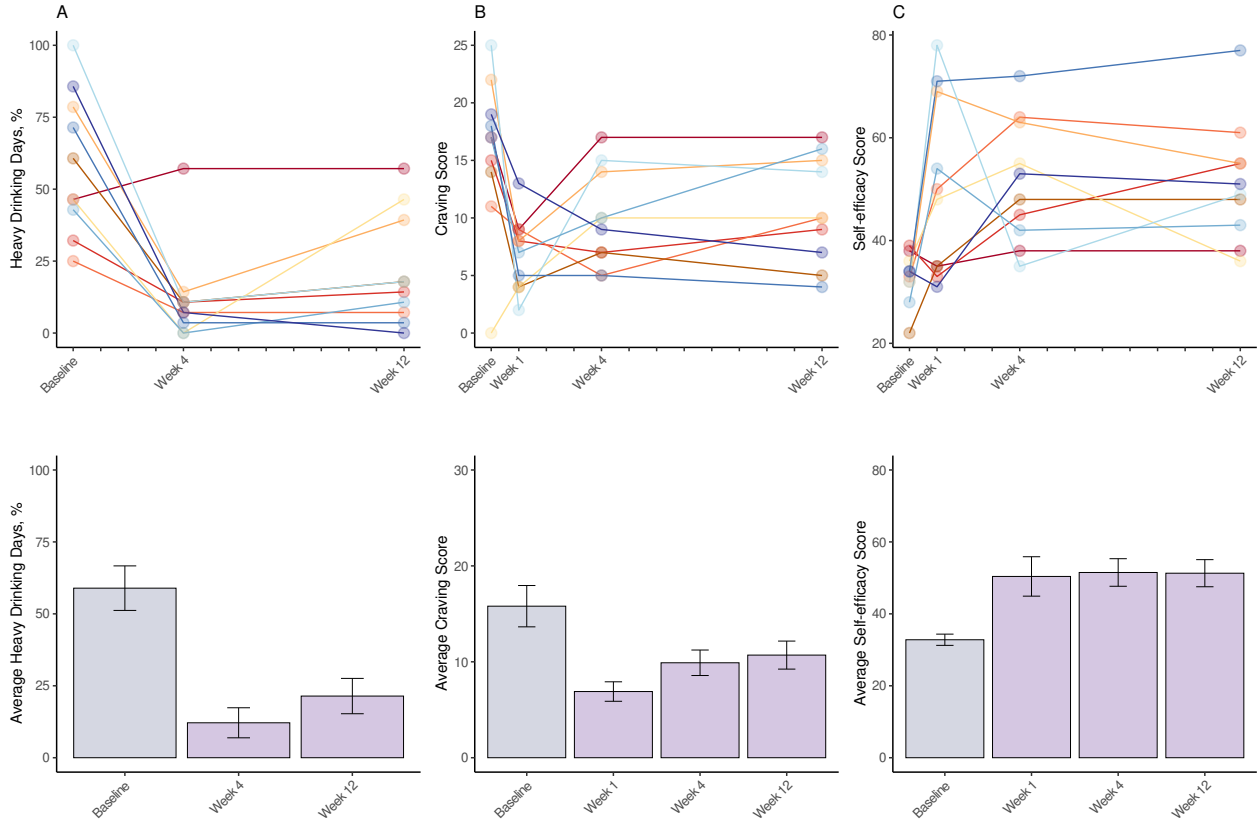


Figure 3 – Individual time courses of clinical outcomes Individual and mean changes in the percentage of heavy drinking days, craving, and self-efficacy throughout the study period of 12 weeks. Spaghetti plots depict color-coded data points for each participant, and bar plots depict means with standard errors.

	Baseline	Week 1	Week 4	Week 12
	median (IQR)	median (IQR)	median (IQR)	median (IQR)
Outcomes				
Percentage of heavy drinking days, %	53.6 (43.7, 76.7)	-	8.9** (4.4, 10.7)	16.07** (8.0, 33.9)
Percentage of drinking days, %	67.9 (49.1, 81.2)	-	32.1* (14.2, 45.5)	48.2** (24.1, 56.2)
Drinks per day, No	5.5 (4.2, 7.5)	-	1.44** (0.4, 1.7)	1.69* (1.0, 3.5)
AUDIT	23.5 (20.2, 27)	-	-	13.5** (9.2, 21.2)
Craving	17 (14.2, 18.7)	7.5** (4.2, 8.7)	9.5** (7.0, 13.0)	10.0 (7.5, 14.7)
Temptation	43.5 (35.5, 51)	28.5* (12.7, 34.7)	17.5** (16.0, 25.0)	25.0** (19.6, 31.5)
Self-efficacy	33.5 (32.0, 35.5)	49* (35.0, 65.2)	50.5* (42.7, 61.0)	50.0*** (44.2, 55.0)
Mindfulness	4.1 (3.6, 4.4)	4.2 (3.7, 4.5)	4.1 (3.4, 4.5)	3.7 (3.4, 4.2)
Psychological flexibility	21.5 (15.2, 24)	19.5 (13.2, 22.5)	13.5 (12.0, 21.0)	14.0 (14.0, 16.5)
Depressive symptoms	14 (12.0, 20.2)	10.5 (9.2, 20.7)	11.5 (8.0, 15.0)	10 (10.0, 25.0)

Table 2 - Summary of clinical outcomes. The percentage of heavy drinking days, percentage of drinking days, and drinks per day were quantified using the Timeline Followback Method, which was used to review the past 28 days. Craving was quantified using the Penn Alcohol Craving Score; Temptation and Self-efficacy were quantified using the Alcohol Abstinence Self-efficacy Scale; Mindfulness was quantified using the Mindfulness Awareness Attention Scale, Psychological flexibility was quantified using the Acceptance and Action Questionnaire, and Depressive symptoms were quantified using the Major Depression Inventory. AUDIT, Alcohol Use Identification Test. Asterisks mark significant differences from baseline as determined by a linear mixed model. P-value < 000.1***, < 0.001 **, < 0.05*.

Discussion

In this open-label study, single-dose psilocybin therapy treatment for AUD showed feasibility, safety, and efficacy in improving drinking outcomes during a 12-week period. Notable variations in the pharmacokinetic values were observed but did not impact the efficacy or the safety.

All participants completed the therapy course, and the safety profile was favorable with no serious adverse events, consistent with the existing literature (Kaminski and Reinert, 2023). Notably, repeated blood sampling for the evaluation of pharmacokinetics was feasible and well-tolerated, even in this vulnerable psychiatric patient population. This aspect is particularly important as it allows future investigations into proposed underlying biological therapeutic mechanisms, including acute effects in neuroplasticity (Ly et al., 2018) and inflammation (Flanagan and Nichols, 2018), which until now have been limited to reports done in healthy subjects.

The time course of subjective drug effects as rated by SDI closely mirrored that of the concentrations of plasma psilocin, perhaps best illustrated in Figure 2 by the profile of participant no 8 (blue color), who exhibited significantly delayed kinetics and consequently delayed dynamics. A tight temporal coupling between subjective drug effects and plasma psilocin is well-documented in healthy subjects (Madsen et al., 2019; Ley et al., 2023; Holze et al., 2023), but here, we demonstrated it for the first time in individuals with AUD. Considerable variation in kinetics and dynamics was observed between participants after a 25 mg dose, with

peak plasma concentrations ranging from 14 to 59 µg/L, consistent with variations seen for all classic psychedelics in healthy subjects (Holze et al., 2024). However, this variance in C_{max} among individuals with AUD (46%) was somewhat greater than in healthy subjects (20%) (Holze et al., 2023) despite the same fixed dose of 25 mg. Variations in kinetics and dynamics may be influenced by several physiological (Dinis-Oliveira, 2017) and psychological factors (Haijen et al., 2018), respectively. Previous studies suggest that major intrinsic pharmacokinetic factors, including weight and age, do not appear to impact the pharmacokinetics of plasma drug concentrations in healthy subjects (Holze et al., 2023) or subjective drug effects in cigarette smokers, in individuals with life-threatening cancers, and those with major depressive disorders (Garcia-Romeu et al., 2021; Spriggs et al., 2023). The present study aligns with these previous findings as we did not find relevant associations between body weight, age, liver and kidney function and plasma concentrations, and subjective drug effects. Of note, the participants in our study had normal liver and kidney function tests and showed no clinical signs of impairments. Moreover, body weight was 60 to 95 kg. Therefore, it remains uncertain whether impaired liver and kidney function or body weight affect the kinetics and dynamics of psilocybin in individuals with AUD. Future studies should also consider other important factors that may alter the metabolic pathway of psilocin, including the enzymes aldehyde dehydrogenase and UDP glucuronosyltransferases (Dinis-Oliveira, 2017) which may be affected by chronic alcohol intake. The lack of strict dietary control in the present study may also have contributed to the variability. Due to sample size limitations, we were unable to ascertain if sex influenced the effects of psilocybin, which should be investigated by future studies.

We observed marked improvements in drinking outcomes, with a large effect size (Cohen's d = 1.02). The effect size was on par with similar research (Bogenschutz et al., 2015) that employed a more extensive intervention, including two dosing sessions. Despite not observing an increase in traits such as mindfulness or psychological flexibility, which have been proposed as therapeutic mechanisms (Madsen et al., 2020; Davis et al., 2020; Doss et al., 2021), the study's findings align with existing data showing a decrease in alcohol craving, temptation and an increase in self-efficacy (Bogenschutz et al., 2022; Bogenschutz et al., 2015), reinforcing the potential benefits of this therapeutic approach. Consistent with the accumulating evidence supporting the notion that the subjective effects play a central role for the long-term efficacy (Ko et al., 2022), our exploratory analysis (see Figure S3) suggests that profound mystical-type experiences were associated with sustained improvements in reducing heavy drinking days. Interestingly, plasma concentrations of psilocin did not appear to be associated with these improvements.

These promising results notwithstanding, the present study has some key limitations, including a small sample size with only two women, lack of a control group, and an open-label design, which prohibits any causal conclusions regarding efficacy. Additionally, selection bias due to self-referral and high expectations from

positive widespread media coverage may have influenced, at least in part, the observed positive effects. Therefore, larger, placebo-controlled single-dose trials, currently underway (NCT05416229), are necessary to establish firm conclusions.

In conclusion, single-dose psilocybin therapy demonstrated feasibility, safety, and efficacy in reducing alcohol use in AUD patients over 12 weeks. Despite notable pharmacokinetic variations, no adverse impact on safety or efficacy was observed, supporting the clinical relevance of a 25 mg fixed dose.

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Author contributions

MEJ conceived the study, was Co-Principal Investigator, wrote and applied for initial regulatory approvals and protocol, conducted screenings and assessments, provided psychological support for participants, oversaw the clinical research coordination, performed the data analysis, and was the primary author of the manuscript. DSS oversaw the intellectual conceptualisation and development of the psychological support model, supervised the psychological support provided for the participants, and reviewed the manuscript. CDM contributed to developing the psychological support model, provided psychological support for the participants, and reviewed the manuscript. EDP oversaw the data acquisition for the pharmacokinetics, contributed to the clinical research coordination, conducted assessments, and reviewed the manuscript. TVV oversaw the statistical analysis and reviewed the manuscript. PMF contributed to the intellectual conceptualisation of the study and reviewed the manuscript. MKKN and SSJ oversaw the analysis of plasma psilocin levels and reviewed the manuscript. NDV contributed to the interpretation of the data and reviewed the manuscript. GMK contributed to the intellectual conceptualisation of the study and reviewed the manuscript. AFJ conceived the study, was the Principal Investigator, secured funding and regulatory approval, provided executive oversight of the study at large, and oversaw the editing of the manuscript.

Statements and declarations

Ethical considerations

All procedures were approved by the Danish Ethics Committee of the Capital Region, Copenhagen, Denmark (H-20043832) and the Danish Medical Agency (2020-000829-55). The study was conducted according to the Declaration of Helsinki and was monitored by an independent study monitor (Good Clinical Practice unit, Copenhagen, Denmark).

Consent to participate

All participants received oral and written information and provided informed consent before being included in the study. Participants did not receive compensation, either monetary or otherwise.

Declaration of conflicting interest

GMK has served as a speaker for Angelini, Abbvie, Cybin, and H. Lundbeck and as an advisor for Sanos, Onsero, Pangea Botanica, Gilgamesh, and Seaport. The remaining authors report no competing interests.

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Data Availability

The datasets generated during the current study are available from the corresponding author upon reasonable request.

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